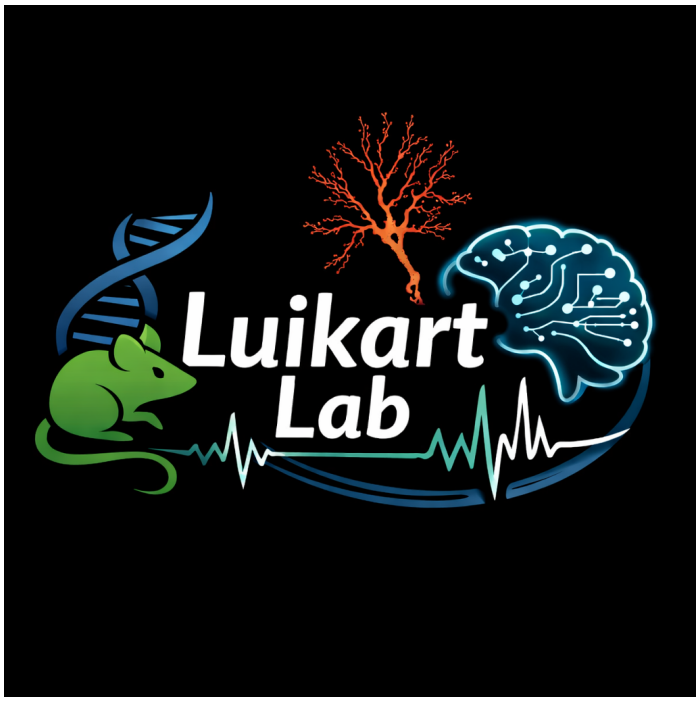


Soma Detector Suite

User Guide · Version 2.2 · Luikart Laboratory

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1. What the suite does

The Soma Detector Suite is three programs that work together to measure the fluorescence of neuronal cell bodies (somata) in confocal images and compare experimental groups statistically. You move through them in order:

- **Soma Detector** — opens a Zeiss **.czi** or **.lsm** image, automatically finds somata, lets you review and correct them, and exports per-cell measurements to Excel.
- **Cre Classifier** — reads those measurements and labels each soma **Cre+**, **Cre**–, or **both** based on its fluorescence.
- **Group Analysis** — pools the classified cells across animals, runs the statistics, and makes publication figures.

A key idea in this version: the information needed to group your data — the **animal ID**, the **mouse genotype**, and what each **channel** is — is entered once in the Soma Detector and travels with the data through every later file. Grouping no longer depends on how files are named.

2. Getting started

- Run **SomaDetectorSuite_Setup_v2.2.exe** and follow the prompts. No Python or other software is required.
- A single **desktop icon** (the orange neuron) opens a folder containing the three programs.
- The **first launch** of each program takes a few seconds while it unpacks; after that it is quick.

- Typical order of use: **Soma Detector** → **Cre Classifier** → **Group Analysis**.

> **Tip.** On a machine downloading the installer for the first time, Windows may show “Windows protected your PC.” Click **More info** → **Run anyway**. This is expected for in-house software that is not code-signed.

3. Soma Detector

Detect, review, and measure somata in one image (or a whole folder).

Open an image and set up the channels

- Click **Open .czi file** to load a .czi or .lsm image. To reopen work you saved earlier, click **Load ROI Set** and choose the matching ***_ROIs.npz** file.
- In **Channel Assignment**, each physical channel is shown by its emission/excitation wavelength (objective, independent of any dye label the microscope wrote). Confirm the three roles — two detection channels and one measure-only channel — and type the real fluorophore or antibody name for each (e.g., GFP-Cre, mCherry, CaV2.1). Click **Apply assignment & re-detect**.
- Enter the **Animal ID** (a unique per-mouse identifier such as CC89603_111720#1) and the **Mouse Genotype**. Both are guessed from the filename and are fully editable; use the **Italic** and **x²** buttons for standard nomenclature (e.g., italic gene, superscript allele).

Detect and curate

- Adjust each channel’s **Sensitivity** and **Outline breadth** sliders and click **Re-detect**. Fine morphological criteria live in the collapsible **Shape Filters** panel.
- Use the slice slider to move through the stack; **position 0 is a Z-projection** showing every accepted outline at once — the fastest way to spot false positives.
- **Click an outline** to reject or restore it (rejected outlines turn grey and are excluded from export).
- Turn on **Force-detect mode** and click a missed soma to add it. Use **Edit ROI shape** to drag an outline’s vertices and fix a boundary.
- The **Min/Max Mask** highlights saturated (hot pink) and empty (dark blue) pixels; **Intensity Scaling** normalizes each plane to correct depth-dependent dimming.

Export

- Click **Export measurements**. This writes the raw and intensity-scaled Excel workbooks, a refiner CSV, and the curated ***_ROIs.npz** — all named after the folder and accumulating one row per soma. The leading columns are **File, AnimalID, Genotype**.
- To process a whole folder at once, click **Batch Process Directory**. Each image must already have a saved ***_ROIs.npz** (the record of your curation), which the batch reuses.

> **Tip.** The animal ID, genotype, and channel names you set here are saved inside the ROI file, so re-opening it — or batch-processing a folder of many animals and genotypes — restores each file’s own correct information.

4. Cre Classifier

Label each soma by its fluorescence phenotype.

- Click **Open** and choose the measurements workbook (usually the ***_intensity_scaled_soma_measurements.xlsx**).
- Confirm which channel is the **Cre** channel (shown in green) and which is the **Counter** channel (shown in red) using the dropdowns.
- Drag the two cutoff sliders. The three scatter plots and the live counts (Cre+ / Cre– / Both / Unclassified) update as you move them. Optionally rename the phenotype groups.
- Click **Export**. This writes **classified_<name>.xlsx** with a standardized **Classification** column, carrying the AnimalID and Genotype columns through unchanged. Cells below both cutoffs are omitted.

5. Group Analysis

Pool across animals, run statistics, and make figures.

- Click **Open file(s)** and choose one or more **classified_*.xlsx** files (use **Add file(s)** to pool more into the same analysis).
- Groups are built from the data, not the filename: the control group is the Cre– cells (labelled **Cre**–), and each knockout group is labelled by its genotype plus Cre-positive status (e.g., *Pten*$^{flx/flx}$ Cre+), rendered in nomenclature on the plots.
- Pick the **Outcome variable** to plot — for example soma **Area**, the mean of your measured channel, or its control-normalized version **Mean <antibody> / Cre**– (the measured, third channel is divided by nearby Cre-negative cells in the same slice).
- Choose which conditions to include, the reference group, an optional z-depth limit, and the Cre– averaging window.
- Click **Run statistical tests**. The reported test is a **Welch t-test on per-animal means** (cell-level and Mann–Whitney tests are shown alongside), Bonferroni-corrected. Use **View full results** and **Export stats to xlsx**, and **Export figure** for a 300-dpi image.
- Need to correct a grouping? **Edit Animal IDs** and **Grouping setup** are pre-filled from the data and let you override any animal or condition.

> **Tip. Click any point on the plot** to identify that cell, then **Open in soma detector** to jump straight to it in the original image. The Soma Detector opens automatically if it is not already running, and its window is brought to the front so you cannot miss the cell's outline blinking.

6. Tips & troubleshooting

- **SmartScreen warning** on a new machine: **More info** → **Run anyway**. Some antivirus may scan the programs on first run; this is normal for unsigned in-house tools.
- **Nothing happens for a few seconds** after double-clicking a tool the first time — it is unpacking. Give it a moment.
- **Click-to-jump** from Group Analysis needs the original image reachable on disk; keep images and their spreadsheets together.
- **Keep files together.** The measurements, classified files, and ROI archives for a dataset are meant to live in the same folder.